NAMAA at DialectSentEval 2026 Subtask 2: From Reranked Sampling to Cross-Backbone Ensembling for Arabic Dialectal Sentiment Swap

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Abstract

Arabic sentiment swap aims to reverse the polarity of a dialectal sentence while preserving its underlying meaning. We present NAMAA's system for DialectSentEval 2026 Subtask 2. Fine-tuning AraT5v2 achieves 79.56 development sentiment-style accuracy, while reranked sampling over eight candidates, using sentiment agreement and content-overlap signals, raises this to 93.57. Cross-backbone ensembling further reaches a development score of 96.78. On the hidden test set, however, this ordering does not transfer: a widened rerank pool with a re-tuned content weight achieves our best official score of 0.7597, whereas the most complex all-source ensemble scores 0.7343. We identify two likely contributors to this development-to-test mismatch: the absence of gold source-polarity labels at test time, which requires classifier-based polarity inference, and train–development sentence duplication that may inflate development performance. These results show that additional generation and ensemble complexity can improve development metrics substantially without guaranteeing better generalization under shifted evaluation conditions.

1 Introduction

Sentiment style transfer aims to reverse the polarity of a sentence while preserving its underlying meaning and relevant content. This task is particularly challenging for Arabic because sentiment is expressed across substantial morphological, lexical, orthographic, and dialectal variation, often alongside code-switching. DialectSentEval 2026 Subtask 2 (Ezzini et al., 2026a) formulates this problem as dialectal Arabic sentiment swap using an extended, dialect-augmented version of the MA'AKS parallel corpus (Mughaus et al., 2026). The task requires systems to balance two competing objectives: reliably reversing sentiment polarity while maintaining content close to the intended rewritten sentence.

We approach the task with a sequence-to-sequence generation framework augmented by candidate reranking and cross-backbone ensembling. Fine-tuned Arabic generation models provide the initial candidate space, after which multiple sampled hypotheses are evaluated using sentiment agreement and content preservation signals. This substantially improves development performance: reranked sampling raises sentiment-style accuracy from 79.56 to 93.57, while cross-backbone ensembling reaches 96.78. However, this development-time advantage does not translate directly to the hidden test set, where our best official submission achieves 0.7597 and a simpler reranking configuration outperforms the most complex ensemble.

Our main contributions are threefold. First, we introduce a reranked-sampling strategy that selects among multiple generated candidates using sentiment and chrF-based content signals, providing the largest improvement over single-output decoding. Second, we investigate cross-backbone ensembling together with non-neural lexical baselines, including antonym substitution, negation editing, and masked-span infilling, to characterize the value of generative model diversity. Third, we analyze the substantial development-to-test mismatch and identify two important factors: the absence of gold source-polarity labels at test time and measurable train-development duplication. These findings suggest that increasing generation and ensemble complexity is not sufficient when evaluation-time conditions differ from those available during development.

2 Related Work

DialectSentEval 2026 Subtask 2 (Ezzini et al., 2026a) formulates Arabic sentiment swap as a controlled generation task: given a dialectal Arabic sentence and its source polarity, the system must produce a sentence with the opposite sentiment while preserving the original semantic content. The

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1436–1440
October 28-29, 2026 ©2026 Association for Computational Linguistics

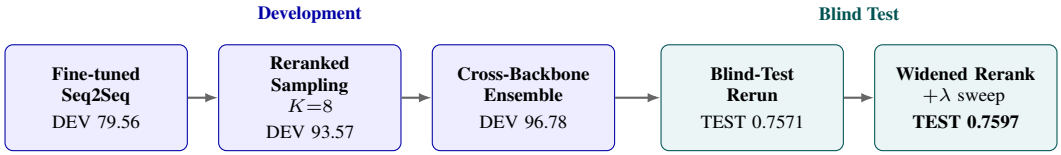


Figure 1: System progression from fine-tuned Seq2Seq models through reranked sampling and cross-backbone ensembling.

task covers four broad dialect groups—Maghrebi, Egyptian, Gulf, and Levantine—although individual examples are not accompanied by explicit dialect labels. Its dataset, *MA'AKS Extended*, expands the original 5,000-pair MSA-only MA'AKS corpus (Mughaus et al., 2026) with dialectal data, yielding 9,647 pairs: 6,752 for training, 964 for development, and 1,931 for blind testing. Training and development examples provide parallel target rewrites, whereas the blind test contains only `id` and `source`.

Evaluation considers both sentiment reversal and content preservation. Sentiment style accuracy is computed with CAMeLBERT-DA-sentiment (Inoue et al., 2021), measuring whether the generated sentence matches the intended opposite polarity. Content preservation is evaluated with BLEU and chrF against the reference rewrite. From CodaBench scorer outputs, we verified that this comparison is performed against the gold target rather than the source sentence. The exact formula used to combine these criteria into the single leaderboard score is not publicly specified; accordingly, we report the official composite scores without inferring their internal weighting and use an identity system as our own development baseline.

Prior Arabic sentiment-analysis research has progressed from lexical and classical machine-learning approaches toward pretrained Transformer models, while robustness across dialects remains challenging (Shi and Agrawal, 2025). Sentiment swap is more closely related to controllable text and style-transfer generation than to conventional sentiment classification. Early style-transfer work demonstrated that textual attributes can be modified while attempting to preserve semantic content (Shen et al., 2017). For generation, sampling and reranking provide an alternative to deterministic beam or greedy decoding, whose mode-seeking behavior may not reflect the broader model distribution (Eikema and Aziz, 2020). Related optimization approaches include REINFORCE (Williams, 1992) and Minimum Risk Training (Shen et al., 2016), which directly optimize non-differentiable sequence-level objectives.

A closely related but separate evaluation, AraSentEval (Ezzini et al., 2026b), uses the original MSA-only MA'AKS data for sentiment swap. Its winning system combines AraBART and AraT5v2 with targeted data augmentation (Hamdy et al., 2026). Although AraSentEval and DialectSentEval share dataset lineage and task framing, they are distinct competitions with different data conditions and should therefore not be compared by directly merging their reported scores.

3 System Overview

Figure 1 summarizes our approach: fine-tuned backbones, reranked sampling, then cross-backbone ensembling, with non-neural lexical baselines for contrast.

3.1 Fine-tuned Sequence-to-Sequence Backbones

We fine-tuned four encoder-decoder backbones on the 6,752 training pairs (AraT5v2-base-1024, AraT5-base, mT5-base, AraBART) under two input formats: a control-code *prefix tag* and an *Arabic instruction prompt*. AraT5v2 with the Arabic prompt (S06a) was strongest at 79.56 development sentiment-style accuracy; AraT5-base trailed at 60.27, mT5-base at 67.12, AraBART at 76.87 as shown in Table 1. A beam width sweep (4→8) moved this by only −0.20, showing decoding width was not the bottleneck.

3.2 Reranked Sampling

Rather than decoding greedily, we sample K=8 candidates (temperature 0.9, top-p 0.95) from the strongest checkpoint, classify each with the scoring classifier, and pick the candidate maximizing $100 \cdot \mathbb{1}[\text{style match}] + \lambda \cdot \text{chrF}(\text{candidate}, \text{source})$, λ=0.3 (reranked against *source*, not the unavailable gold target, to stay deployable). This raised development accuracy from 79.56 to 93.57, a 14-point jump for no extra training cost. It was the biggest lever we found.

3.3 Cross-Backbone Ensembling

Reranked sampling on a second backbone (AraBART), combined with the AraT5v2 pool and

a beam-decoded source, reranked jointly: diversity compounded a 3-source ensemble to 88.28, a 5-source ensemble to 90.04, and our "grand ensemble" (AraT5v2 reranked + AraBART reranked + beam control) to 96.78, our development ceiling (Table 1).

Non-neural lexical baselines. Three training-free contrastive baselines: a mined *antonym substitution table* (48.76, word-aligned majority replacements, 860/964 rows covered); *rule-based negation insertion/removal* (23.34, barely above identity); and *masked-span substitution* via occlusion saliency + BERT fill-mask (53.53–58.40), extended to a tag-then-infill variant (57.78). All trail the fine-tuned backbones by a wide margin. Figure 2 shows why: most of the 560 failures need more than a single-word fix, consistent with reranked generation's much larger gain (Section 3.2).

4 Experimental Setup

Data. Training (6,752) and development (964) are fully parallel; the blind test file (1,931) carries only `id` and `source`. Since our pipeline needs the current polarity to flip, we infer `source_polarity` on test via the scoring classifier on the source, remapping Neutral to the higher-scoring polarity (9.4% of rows); this circular step is one cause of the development/test gap (Section 6).

Training and decoding. Fine-tuned with the Hugging Face `Seq2SeqTrainer` (Wolf et al., 2020): batch size 16 (32 for the earliest run), 8 epochs, lr 3×10^{-4}, 10% warmup, early stopping (patience 3) on development style accuracy, max length 96, beam width 4. Reranked sampling: temperature 0.9, top-p 0.95, K=8, λ=0.3 (Section 3.2).

Reproducibility of test-phase submissions. Ten systems were submitted for the blind test phase (named in Table 2): fixed/widened reranked-ensemble reruns (1, 5), added candidate sources (2, 9), iterative self-refinement (4), a Minimum-Risk-Training fine-tune (Shen et al., 2016) against the reranking reward (6), a λ-bracketing probe (7, 8), and a grand ensemble (10). Only system (5)'s lineage is fully locally reproducible; the rest were submitted directly to CodaBench, and we report their official scores (see Limitations).

Metrics and tools. We evaluate content preservation with BLEU using `sacrebleu` (Post, 2018) and chrF (Popović, 2015). Model training and inference use Hugging Face `transformers` (Wolf et al., 2020), including AraT5v2 (Elmadany et al., 2023; Nagoudi et al., 2022), AraBART (Kamal Eddine et al., 2022), mT5 (Xue et al., 2021), and CAMeLBERT-DA-sentiment (Inoue et al., 2021). Official blind-test evaluation is performed through CodaBench (Xu et al., 2022). Our code[1] and released checkpoints[2] are publicly available.

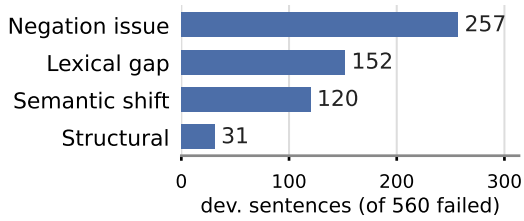


Figure 2: Lexicon+negation failure taxonomy (dev).

System	Dev Style Acc.
Baseline	
Identity (no-op)	22.20
Non-neural lexical baselines	
Negation insertion/removal (S15)	23.34
Mined antonym table (S16)	48.76
Masked-span, AraBERTv2 MLM (S14b)	53.53
Masked-span, CAMeLBERT-mix MLM (S14a)	58.40
Tag-then-infill (S19)	57.78
Single fine-tuned backbones	
AraT5-base, prefix tag (S03)	60.27
mT5-base, prefix tag (S04)	67.12
AraBART, prefix tag (S05)	76.87
AraT5v2, prefix tag, bs=32 (S01)	76.76
AraT5v2, Arabic prompt (S02)	78.11
AraT5v2, Arabic prompt (S06a)	79.56
+ beam sweep, beams=8 (S06b)	79.36
Reranked sampling	
S06a + rerank, K=8, λ=0.3 (S09)	93.57
AraBART + rerank (E4)	91.08
Cross-backbone ensembles	
3-source: S06a, S02, S05 (S13)	88.28
5-source: S13 + S01, S04 (E2)	90.04
3-source: S09, S05, S02 (E3)	95.12
Grand: S09, E4, S02 (E5)	**96.78**

Table 1: Development-set Sentiment Style Accuracy; full run logs at the code repository (Section 4). Bold marks the highest score; Section 6 explains why this does not predict hidden-set performance.

5 Results

5.1 Development Results

Table 1 reports our full development log, 19 configurations spanning 22.20 to 96.78 sentiment-style accuracy (Section 6 explains why this does not transfer to test).

5.2 Official Test Results

Table 2 reports the official CodaBench composite score for all ten blind-test submissions (Section 4); our best submission, system (5), a widened rerank pool with a re-tuned content weight, reached 0.7597.

[1] https://github.com/NAMAA-ORG/DialectSentEval-Arabic-Sentiment-Swapp

[2] https://huggingface.co/collections/HassanB4/dialectsenteval-2026-arabic-sentiment-swap

System	Test Score
E5 grand ensemble, direct carry-over	0.7571
(1) Fixed E5 rerun	0.7582
(2) + masked-span 4th member	0.7364
(4) Iterative self-refinement	0.7514
(5) Widened pool + λ sweep	**0.7597**
(6) Minimum-Risk-Training fine-tune	0.7193
(7) Style-weighted rerank ($\lambda \to 0$)	0.7582
(8) Content-weighted rerank (λ large)	0.7582
(9) + antonym-table 4th member	0.7431
(10) Grand ensemble (all sources)	0.7343

Table 2: Official CodaBench Sentiment Swap composite scores for the blind test phase.

5.3 Error Analysis

Systems (7) and (8) rerank the identical candidate pool with λ pushed toward opposite extremes (near-pure style vs. content) and score identically at 0.7582, matching system (1)'s unmodified λ=0.3: within this candidate pool, the leaderboard composite appears insensitive to this reranking weight, suggesting the pool's diversity, not the rerank weighting, binds this test distribution. System (6), the only submission that fine-tunes further (Minimum-Risk-Training against the reranking reward), scores lowest among the gradient-based variants (0.7193), below even the identity-style rerank baselines, indicating this reward does not transfer cleanly into a training objective at this scale. Most strikingly, system (10), the richest ensemble folding in every candidate source we built, scores 0.7343, below every system except (6) and (2), and well below system (1)'s much simpler fixed rerun (0.7582): the diversity-via-ensembling pattern that reliably improved development sentiment-style accuracy (Section 3.3) does not reproduce on test, where the reranker chooses among noisier, inferred-polarity candidates rather than clean, gold-polarity ones.

6 Discussion

The development log as shown in Table 1 is not predictive of test-set ranking, not from simple overfitting: no system trains on or copies development gold targets, since reranking deliberately scores against *source*, not *target*, avoiding a non-deployable oracle as shown in Section 3.2. Two factors likely contribute to the gap. First, every system needs a gold `source_polarity` label to know which direction to flip; development gives this as ground truth, but the blind test file does not, forcing a circular classifier-based inference as referred to in Section 4 that likely drives why the development-winning ensemble's direct carry-over (0.7571) sits far below its 96.78 development score. Second, using EDA on the released source text, we found 123/964 (12.8%) development source sentences exactly duplicate training sentences, plus substantial within-split duplication (17% training, 7% development), inflating the development ceiling for anything that indirectly relies on having seen a near-identical training example. Development ranking is accordingly a poor guide to generalization: the simplest carried-over system (5) outperforms the most elaborate one (10) on the split that matters.

7 Conclusion

We presented a reranked-sampling and cross-backbone ensemble for Arabic dialectal sentiment swap. Reranking was the strongest development-time improvement, while the best hidden-test result came from a simpler widened rerank pool, reaching 0.7597. The development-to-test gap is likely influenced by missing test-time polarity labels and train-development duplication, while the unpublished leaderboard formula and reliance on the same sentiment classifier for reranking and evaluation further limit interpretation. Future work should prioritize robust polarity inference and cleaner evaluation over additional ensemble complexity.

Ethical Considerations

Automated sentiment-polarity inversion is dual-use: it could fabricate opposite-sentiment reviews at scale; we release it for research only. Our EDA found uneven dialect-marker representation (Levantine 20.5% of rows, Maghrebi 2.2%), so performance may vary across unlabeled dialects.

Acknowledgments

We thank Tuwaiq Academy for its continued support of this work. We also thank the DialectSentEval 2026 organizers.

References

Bryan Eikema and Wilker Aziz. 2020. Is MAP decoding all you need? the inadequacy of the mode in neural machine translation. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 4506–4520, Barcelona, Spain (Online). International Committee on Computational Linguistics. Best paper award; motivates sampling-and-reranking over mode-seeking (beam) decoding, which we found empirically to be our single largest development-set lever.

AbdelRahim Elmadany, El Moatez Billah Nagoudi, and Muhammad Abdul-Mageed. 2023. Octopus: A multitask model and toolkit for Arabic natural language generation. In *Proceedings of ArabicNLP 2023*,

pages 232–243, Singapore (Hybrid). Association for Computational Linguistics. Introduces AraT5v2, the checkpoint family we fine-tune (UBC-NLP/AraT5v2-base-1024).

Saad Ezzini, Shadi Abudalfa, Maram Alharbi, Salmane Chafik, Hind Alatawi, Mo El-Haj, Ahmed Abdelali, Osamah Alnahari, and Salima Lamsiyah. 2026a. Dialectsenteval 2026: Arabic dialect sentiment analysis and swapping shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Official organizer-provided citation, received by email 2026-08-09.

Saad Ezzini, Shadi Abudalfa, Maram Alharbi, Salmane Chafik, Hamzah Luqman, Mo El-Haj, Paul Rayson, Samaher Alghamdi, and Reem Alotaibi. 2026b. AraSentEval: A shared task on sentiment analysis and swapping in Arabic. In *Proceedings of the 7th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT7), co-located with the Language Resources and Evaluation Conference (LREC 2026)*, Palma, Mallorca, Spain. Official organizer-provided citation for the related OSACT7/LREC 2026 task, received by email 2026-08-16.

Yumna Hamdy, Mohab ElDamhougy, Yomna Eid, and Ensaf Hussein. 2026. A comparative study of Arabic sentiment swap models for AraSentEval 2026. In *The 7th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT7) with 5 Shared Tasks*, pages 269–273. Winning system for the related, separately-run AraSentEval 2026 Subtask 2 at OSACT7/LREC 2026.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Moussa Kamal Eddine, Nadi Tomeh, Nizar Habash, Joseph Le Roux, and Michalis Vazirgiannis. 2022. AraBART: a pretrained Arabic sequence-to-sequence model for abstractive summarization. In *Proceedings of the The Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 31–42, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Raed Mughaus, Shadi Abudalfa, Hamzah Luqman, Fahad Abdu, Mohammed AlAli, Nawaf Al-Dowayan, and Ahmed Abdelali. 2026. Ma'aks: manually-curated parallel dataset for Arabic text sentiment swap. *Language Resources and Evaluation*, 60(1):1.

El Moatez Billah Nagoudi, AbdelRahim Elmadany, and Muhammad Abdul-Mageed. 2022. AraT5: Text-to-text transformers for Arabic language generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pages 628–647, Dublin, Ireland. Association for Computational Linguistics.

Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, Lisbon, Portugal. Association for Computational Linguistics.

Matt Post. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Brussels, Belgium. Association for Computational Linguistics.

Shiqi Shen, Yong Cheng, Zhongjun He, Wei He, Hua Wu, Maosong Sun, and Yang Liu. 2016. Minimum risk training for neural machine translation. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1683–1692, Berlin, Germany. Association for Computational Linguistics. Minimum Risk Training; the direct precedent for our REINFORCE-with-baseline fine-tune (System 6).

Tianxiao Shen, Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2017. Style transfer from non-parallel text by cross-alignment. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*.

Zhiqiang Shi and Ruchit Agrawal. 2025. A comprehensive survey of contemporary Arabic sentiment analysis: Methods, challenges, and future directions. *arXiv preprint arXiv:2502.03827*. Accepted to NAACL 2025.

Ronald J. Williams. 1992. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Zhen Xu, Sergio Escalera, Adrien Pavão, Magali Richard, Wei-Wei Tu, Quanming Yao, Huan Zhao, and Isabelle Guyon. 2022. Codabench: Flexible, easy-to-use, and reproducible meta-benchmark platform. *Patterns*, 3(7):100543.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498, Online. Association for Computational Linguistics.